

Making Optimal Political Choices

Ben Ansell

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1 Partial Derivatives

In the first lecture of the class you will learn how to take simple derivatives such as $y = x^3 + 2x^2 + x + 4$. But in most applications we have lots of variables of interest. For example, when we look at the determinants of successful democratization we examine not only economic growth but the degree of inequality and the behavior of the military. And these different forces might depend on each other: for example, inequality might matter only in times of low economic growth. We would be in trouble if differentiation only worked in cases with a single variable of interest.

Fortunately, partial differentiation allows us to extract the effects of one particular variable at a time. In essence, we will be *holding all the other variables constant* while just examining one particular variable of interest. When social scientists use the phrase *ceteris paribus*, or ‘all things being equal’, this is what they typically mean.

Let us consider an example from Jeff Gill, (2006) p. 238. Gill is looking at Jerome Gordon’s (1969) analysis of socio-economic status, which he labels as *SES*. Status depends on ‘all terms not included in the model’ (!) *A*, education *E*, income *I*, and employment *M*. Note that these terms are all multiplicative, that is, the impact of your education on your status depends on whether you earn a lot of money and whether you are employed. Unemployed but highly educated and wealthy people have zero status in this setup! But excluding this perversity, this set-up is quite sensible. We will look at an alternative in a second. But first let’s replicate Gill’s original equation:

$$SES = AE^b I^c M^d$$

Now, suppose we want to know what the impact of an extra year of education is on status, controlling for your income and employment (let's assume that these are independent of education, which we know they are not). We take a partial derivative, which uses the funny 'd' sign ∂ .

$$\frac{\partial SES}{\partial E} = bAE^{b-1}I^cM^d$$

Notice that the only change relates to E and b . Since other stuff, income, and employment are not changing we can treat them as a *constant* AI^cM^d . The exponent b comes down from E , as we are used to. Whenever you have this kind of multiplicative arrangement, often called a 'Cobb Douglas' function - more in a bit on this - you should see if you can simply rephrase the RHS of the equation using the original pre-derivative equation.

$$\frac{\partial SES}{\partial E} = b \frac{SES}{E}$$

Now we see that the effect of another year of education on status can be thought of as b multiplied by the current level of status per year of education - whatever that means..! Looks simpler though, and this helps us use the new form in further equations.

Before we move on, consider what the partial would have looked like if status was additive not multiplicative. In our new setup, education has an effect on status *regardless* of whether you are employed or earn any money. Good news for graduate students.

$$SES = A + E^b + I^c + M^d$$

$$\frac{\partial SES}{\partial E} = bE^{b-1}$$

What has happened to income, employment, and 'other stuff'? Well, they don't matter now in determining the effect of education on status. They simply drop out because they don't have an E term attached to them. This looks more like the kinds of basic regressions you will do when you start statistical work, where every variable matters only in itself, unless it is explicitly multiplied with another. And obviously this makes partial derivatives much easier. Let's say we have a model of voting turnout, where we think income, education, and race matter, and do not depend on each other. Moreover, we think each variable matters linearly - that is,

an extra year of education will have the same impact on voting regardless of whether it is your first or twenty-first. Then we can set up a classic regression equation:

$$T = \alpha_0 + \beta_1 I + \beta_2 E + \beta_3 R + \varepsilon$$

When we take the derivative of turnout with respect to education we find that everything drops out except β_2 , which is estimated to be the simple effect of one year of education on turnout. What if education and race interrelate, for example, education matters more for blacks than whites in terms of turnout? Then we get, assuming $R = 1$ means black,

$$T = \alpha_0 + \beta_1 I + \beta_2 E + \beta_3 R + \beta_4 ER + \varepsilon$$

Now if we take the partial derivative of this with respect to education we get

$$\frac{\partial T}{\partial E} = \beta_2 + \beta_4 R$$

Now, as suggested, the impact of education on income is also dependent on race. What if we think that income is just a function $g()$ of education? That is, $i = g(e)$. We can stipulate a new turnout equation, where only income and education matter, as follows:

$$t = e^\alpha + i(e)^\beta$$

If we take the partial derivative of this equation we now need to look at the *direct* and *indirect* effects of education:

$$\frac{\partial t}{\partial e} = \alpha e^{\alpha-1} + \beta i^{\beta-1} \frac{\partial i}{\partial e}$$

We finish our discussion of partial derivatives with an aside on the famous Cobb-Douglas equation beloved by economists. Let's assume, as economists are wont to do, that to produce a can of Coke we need some capital stock K , for example, a bottle plant, and labor L . Each input is useful but only with diminishing marginal returns. Accordingly each has an exponent, respectively α and $1 - \alpha$, where $\alpha < 1$. This gives us the Cobb-Douglas formula:

$$Y = K^\alpha L^{1-\alpha}$$

Note that in the Cobb-Douglas equation, adding either labor or capital on their own has diminishing marginal returns because both α and β are less than one. However, the Cobb Douglas function as a whole displays constant returns - that is, if we double *both* capital and labor we will double output. Let t be any constant, including 2 if we want to double inputs.

$$Y' = (tK)^\alpha (tL)^{1-\alpha} = t^\alpha t^{1-\alpha} K^\alpha L^{1-\alpha} = tK^\alpha L^{1-\alpha} = tY$$

Let's take derivatives of output with respect to capital and labor

$$\frac{\partial Y}{\partial K} = \alpha K^{\alpha-1} L^{1-\alpha} = \alpha \frac{Y}{K}$$

$$\frac{\partial Y}{\partial L} = (1 - \alpha) K^\alpha L^{-\alpha} = (1 - \alpha) \frac{Y}{L}$$

Note that these derivatives are easy to interpret. Let's concentrate on the second one. The effect on output of increasing labor input by one worker is output per worker, Y/L , multiplied by $1 - \alpha$. So output increases by less than the average output per worker and the size of the increase depends on the size of α - the smaller the better. Now that we have established these derivatives we can examine how *percentage changes* in labor input affect percentage changes in output. After all, maybe we don't care about how adding one worker affects things. Maybe we want to know what happens if we increase the workforce by ten percent or doubling it. Hopefully you remember how to figure out what a percentage change is from high school. If, like me, you completely forgot it, here's the basic equation.

$$\% \text{ Change} = \frac{\Delta Y}{Y} * 100$$

So we simply take the change in Y , divide it by the previous level of Y , and then multiply by 100. Hmm... this looks familiar. After all, we have been just looking at the change in Y following a change in K . The *elasticity* ϵ of a function is the percentage change in y given a percentage change in x :

$$\epsilon = \frac{\% \text{ Change in } y}{\% \text{ Change in } x} = \frac{\frac{\partial y}{y}}{\frac{\partial x}{x}} = \frac{\partial y}{\partial x} \frac{x}{y}$$

An elasticity of one would imply that doubling x leads to a doubling in y . An elasticity of zero implies no effect. A negative elasticity implies that

increasing x decreases y . And an elasticity greater than one implies that doubling x leads to a more than doubling in y .

We can apply our analysis of elasticity to show what the impact of doubling capital is on output.

$$\epsilon_{Y,K} = \frac{\partial Y}{\partial K} \frac{K}{Y} = \alpha \frac{Y}{K} \frac{K}{Y} = \alpha$$

So the elasticity of output with respect to capital is simply α , which we know is less than one. If $\alpha = 1/2$ then doubling capital investment increases output by fifty percent. Likewise, doubling labor input increases output by $100 * (1 - \alpha)$.

We finish this (long) section on partial derivatives with an analysis of *implicit differentiation*. What if we want to figure the effect of increasing capital input on labor input for a *fixed* level of output? In other words, if we double capital inputs how many workers can we fire? The basic equation for implicit differentiation, using our current variables as examples, is:

$$\frac{\partial L}{\partial K} = - \frac{\frac{\partial Y}{\partial K}}{\frac{\partial Y}{\partial L}} = - \frac{\alpha \frac{Y}{K}}{(1 - \alpha) \frac{Y}{L}} = - \frac{\alpha L}{(1 - \alpha) K}$$

Now, we know how to figure out the elasticity, so let's plug this equation into the elasticity formula:

$$\epsilon_{L,K} = \frac{\partial L}{\partial K} \frac{K}{L} = - \frac{\alpha L}{(1 - \alpha) K} \frac{K}{L} = - \frac{\alpha}{1 - \alpha}$$

So we see that the elasticity is negative and that it depends on the balance between α and $1 - \alpha$. In particular, if $\alpha = 1/2$ we can see that the elasticity will be -1 . Thus increasing capital by ten percent, reduces labor requirements by ten percent.

2 Second Derivatives and Cross Derivatives

So far we have analyzed each partial derivative in turn. But what if we take a derivative of a derivative? Well, surprisingly enough we get a *second derivative*. Why would we do this? The derivative gives us the instantaneous rate of change in a variable. But perhaps we also want to know how this slope is changing around this point. Is it getting ever steeper, or

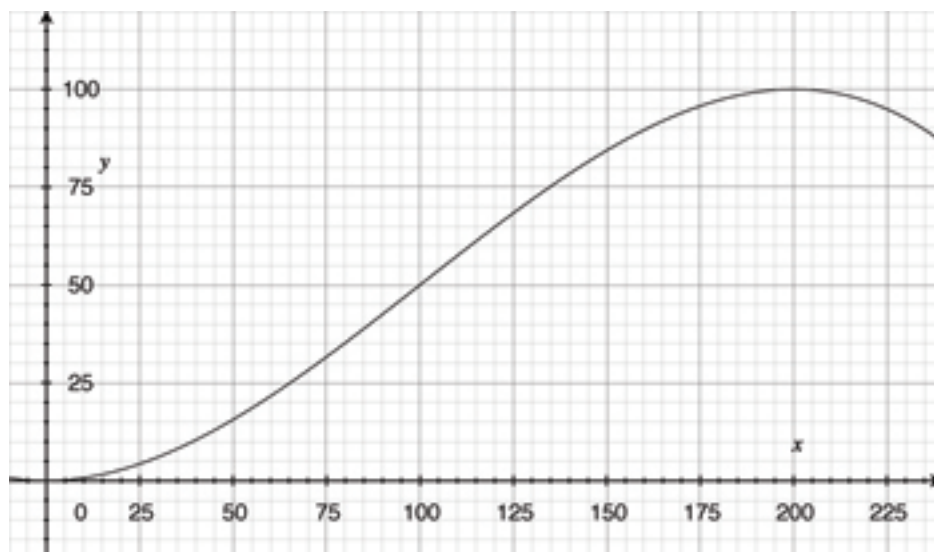


Figure 1: A Car's Journey

is it flattening out. Obviously, this doesn't make sense with a linear equation. A line is a line - it doesn't get steeper or shallower because it never curves. But when we look at a quadratic equation, we might want to know whether the rate of increase is accelerating or decelerating.

The classic example of a second derivative is, indeed, acceleration. If speed is the first derivative of an equation where x is time and y is distance - think about it: miles per hour is the change in y (miles) over the change in x (hours) - then acceleration can be thought of as the second derivative. If a car is speeding up, the line will become steeper and steeper. As it slows down the line becomes flatter, until it stops and the line is totally flat. Thus as the car accelerates the second derivative will be positive, if it maintains constant speed the second derivative equals zero, and if the car slows down the second derivative will be negative.

Figure 1 shows an example of a car accelerating and decelerating. The x -axis is time in minutes and the y -axis is distance in miles. We see the car accelerate and decelerate on its journey in a smooth manner. After 200 minutes the car has travelled 100 miles, so we can say that the average speed of the car is 30mph. However, it is clear that the car only travels at that exact speed for moments during the journey. Note also that after 200 minutes the car apparently starts to reverse!

Let us first consider how fast the car is traveling at any one moment. The equation that produced Figure 1 is the simple function

$$y = \frac{3}{400}x^2 - \frac{1}{40000}x^3$$

We can take the first derivative of this function to find the speed the car is traveling at for any given minute.

$$\frac{\partial y}{\partial x} = \frac{6}{400}x - \frac{3}{40000}x^2$$

And we can further differentiate this expression to find the acceleration or deceleration of the car at any minute.

$$\frac{\partial^2 y}{\partial x^2} = \frac{6}{400} - \frac{6}{40000}x = 0.015(1 - \frac{x}{100})$$

OK, say what does this tell us? Well, we can examine the car's speed and acceleration for various points in time. After 25 minutes the speed is 0.33 miles per minute, or 19.7 miles per hour. The car is accelerating at this point with the second derivative equaling 0.0113, which means that the car is increasing its speed by around 0.68 miles per hour every minute. After 50 miles, the car is going at 0.56 miles per minute, or 33.6 miles per hour - just above its average speed for the trip. Its acceleration, by contrast has slowed to 0.007, so it is gaining 0.456 miles per hour every minute. At 100 minutes, the car is 'haring' along at .75 miles per minute, or 45 miles per hour! Phew... But, let's look at its second derivative. Plug 100 into the last expression above for the second derivative and we see that the equation cancels to zero. So the car is neither accelerating nor decelerating. It is moving at a constant speed.

As we go on, to 150 minutes we see that the car has slowed down considerably, back to 33.6 miles per hour again like the case after 50 minutes. But unlike the case at 50 minutes, when we examine the second derivative we see that it is *losing* speed at 0.46 miles per minute. The second derivative is now *negative*. Finally, at 200 minutes we find that the car has stopped - the first derivative is zero - but with the current equations, we would predict the car to start reversing, since its second derivative is -0.015, and thus the curve must start bending downwards.

We write a second derivative either as $f''(x)$ or as $\frac{\partial^2 y}{\partial x^2}$, where the second expression makes clear the mathematical operation at hand: we are simply

taking the derivative of the first derivative. We typically refer to a curve as concave, quasiconcave, convex, or quasiconvex depending on the sign of the second derivative. A concave curve, $f''(x) < 0$, is always decreasing in rate, it curves downwards and looks like a bump. A quasiconcave curve, $f''(x) \leq 0$, may be always decreasing in rate or straight but it may never start to increase in rate. A convex curve, $f''(x) > 0$, is always accelerating, thus it looks like a bowl - if the slope is negative it is starting to flatten out, if the slope is positive it is climbing faster and faster. Finally, a quasiconvex curve, $f''(x) \geq 0$, is either increasing or straight, never decreasing in slope. Let us now examine two very simple curves: x rooted and x squared.

$$f(x) = x^2, \quad f'(x) = 2x, \quad f''(x) = 2 > 0$$

$$f(x) = \sqrt{x}, \quad f'(x) = .5x^{-1/2}, \quad f''(x) = -.25x^{-3/2} < 0$$

Notice that x squared is always convex, thus we could use it to examine increasing returns when x is positive, whereas x rooted is always concave and thus has diminishing marginal returns. These types of functions are often useful if you are trying to model political and economic behavior with increasing or decreasing returns, although typically we use a parameter, like α , and set it greater or less than one.

The operation we performed on a first derivative to find the second derivative can also be employed with multiple variables, in which case we obtain a *cross-derivative*. Let's examine the equation given in Gill (p.240): $z = 3x^3y^2$. The first derivatives are easy to take: $\frac{\partial z}{\partial x} = 9x^2y^2$ and $\frac{\partial z}{\partial y} = 6x^3y$.

In each case, we are holding the other variable constant to figure out the effect of one independent variable with respect to the dependent variable. But we can see that this effect *is* dependent on the variable we are holding constant. That is, the effect of x on z depends on the level of y . How does this effect depend on y ? We can find out simply by taking the derivative of the effect with respect of y . As you can see, explaining this is not that intuitive! But the equation is:

$$\frac{(\frac{\partial z}{\partial x})}{\partial y} = \frac{\partial^2 z}{\partial x \partial y} = 18x^2y$$

What if we screw up and misread the equation, doing the cross-derivative in the wrong order? No problem. Taking the derivative of the effect of x

on z with respect to y is the same as taking the derivative of the effect of y on z with respect to x (phew...).

$$\frac{(\frac{\partial z}{\partial y})}{\partial x} = \frac{\partial^2 z}{\partial x \partial y} = 18x^2y$$

Let's finish with taking the cross-derivatives of the Cobb-Douglas function from earlier. Recall that $\frac{\partial Y}{\partial K} = \alpha K^{\alpha-1} L^{1-\alpha}$ and $\frac{\partial Y}{\partial L} = (1 - \alpha) K^\alpha L^{-\alpha}$. Take the cross-derivative - remember, it works both ways - to get:

$$\frac{\partial^2 Y}{\partial K \partial L} = (1 - \alpha) \alpha K^{\alpha-1} L^{-\alpha} > 0$$

So we see that the two inputs are *complements* not substitutes, since adding more capital increases the marginal product of labor and vice versa. More generally, variables are complements when their cross-derivative is positive and are substitutes when their cross-derivative is negative. Note also that each input on its own had *diminishing marginal returns* because their second derivatives are negative.

$$\frac{\partial^2 Y}{\partial K^2} = (\alpha - 1) \alpha K^{\alpha-2} L^{1-\alpha} < 0 \quad \frac{\partial^2 Y}{\partial L^2} = -\alpha(1 - \alpha) K^\alpha L^{-\alpha-1} < 0$$

3 Finding Maxima and Minima

What can we do with second derivatives, other than talk about the overall shape of a curve? Well, it turns out they have a very useful function, in that they can tell us if we are at the top of a bump or the bottom of a dip. In other words, they allow us to figure out if we are *maximizing* or *minimizing* a function. In your last class, you saw that if the first derivative equals zero, a slope must be flat. But we don't know if we are at the bottom, top, or in an *inflection point*, where the curve rests before changing direction.

Think back now to the definitions of convexity and concavity in the previous section. If the curve is always increasing in its slope and we have a zero first derivative, then it must be the case that it has previously been a negative slope and is turning positive. After all, if the slope was already positive, we know that since it is increasing in slope forever it can never again be flat. So, it must be the case that a zero first derivative combined with a positive second derivative represents a minimum.

Similarly, if we have a slope that is always decelerating, we must have a concave function - a hump - which implies that any point with a first derivative equal to zero must be a maximum. After all, when a concave function has a positive slope it is decelerating to zero, thus it is reaching a peak.

What if the first derivative is zero and so is the second derivative? At this point we know not only that the line is flat but also that it is changing between accelerating and decelerating (the order being unimportant). This is an inflection point, where a convex curve becomes concave (downward sloping) or a concave curve becomes convex (upward sloping).

The most obvious example is x^3 , which has a second derivative of $6x$. At $x = 0$, the second derivative also equals zero. Yet the first derivative of x^3 is $3x^2$ and must *always* be positive. So, this cannot be a minimum or a maximum, since the slope never becomes negative. Instead, we have an inflection point, where a concave curve becomes convex. Before $x = 0$, x^3 is slowing down but once x becomes positive the function accelerates away.

But let's stop for a minute. Having a zero second derivative is a necessary but not sufficient condition of being an inflection point. For example, the equation $y = 2x$ has a second derivative of zero but no inflection points. To be an inflection point, the second derivative must be positive on one side of the point and negative on the other. It may be necessary to evaluate these points to find out - you may not know just from looking at the second derivative itself.

Furthermore, an inflection point may occur *anywhere* the slope is straight. Look at the equation on Gill (p.244): $f(x) = (x^3 - 15x^2 + 60x + 30)/15$. Here we have a case where there is an inflection point when the first derivative is *not* zero. Take the first derivative: $f'(x) = (3x^2 - 30x + 60)/15$ and the second derivative $f''(x) = (6x - 30)/15$. Note that this latter equation equals zero when $x = 5$ but at this point the first derivative is minus one. So this is neither a maximum nor a minimum but it is an inflection point because at $x = 4$ the second derivative is negative and at $x = 6$ it is positive.

Let's now turn back to maxima and minima. We have seen that the sign of the second derivative determines whether a slope is concave or convex and hence whether a point with a zero first derivative is a maximum or minimum. But if we have a really excitable curve it might snake up and down producing many little bumps and dips. How do we know where the *true* top and bottom of the snake lie?

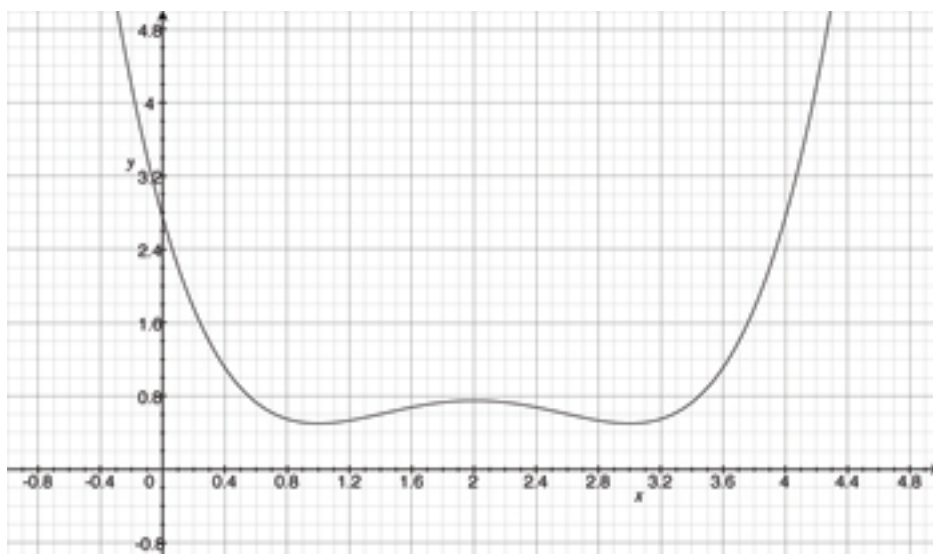


Figure 2: Local and Global Maxima and Minima

This is the difference between *global* and *local* (or *relative*) maxima and minima. Equations that are complex polynomials may have several points at which the first derivative is zero. Look at the equation on page 245 of Gill: $f(x) = \frac{1}{4}x^4 - 2x^3 + \frac{11}{2}x^2 - 6x + \frac{11}{4}$. This has the first derivative: $x^3 - 6x^2 + 11x - 6$. By factoring this equation into $(x-1)(x-2)(x-3)$, Gill shows that there are three points at which the first derivative equals zero: $(1, 2, 3)$.

So the equation is flat in three places, which means that one must be a relative maximum and one a relative minimum (the other could be a max, min, or inflection point). To ascertain which are which we take the second derivative: $f''(x) = 3x^2 - 12x + 11$ and examine this by plugging in $x = (1, 2, 3)$. At $x = 1$, the second derivative evaluates to 2, so this is a local minimum. At $x = 2$ we get a second derivative of -1, a local maximum. Finally, at $x = 3$, we get a second derivative of 2, another local minimum. So we immediately know that one of the local minima is a global minimum because the two minima surround the maximum - there is nowhere else that the curve can dip down.

Which is the lowest minimum? We go back to the original equation and plug in 1 and 3 and find that at both $y = 0.5$. So they are both global minima. What about $x = 2$? Is this the global maximum? We evaluate it to

find that at $x = 2$ $y = .75$. The only other places where y could be higher must be less than one or greater than three. Let's try zero, since that's easy. At $x = 0$, $y = 11/4 > .75$. So $x = 2$ cannot be a global maximum. Where is the global maximum? Let's evaluate the second derivative at $x = 0$ - it equals 11, which means the curve is convex and thus as we go further left on the x -axis the curve must be getting more and more negatively sloped, with y rising to infinity. The same applies at $x = 4$, where the second derivative is also 11. The slope here is becoming more and more positive as we move rightwards along the x -axis up to infinity. So there are no global maxima save ∞ and $-\infty$. Figure 2 shows the equation.t

4 Constrained Optimization

So far we have just looked at the global and local maxima and minima of single equations. What if we, instead, choose to limit the maxima and minima to a *constrained* region? For many political scientists, psychologists, and economists, studying a world of institutional constraints, cognitive limitations, or limited resources, this is a much more useful way of analyzing social behavior.

Constrained optimization is relatively straightforward and is typically approached in two ways. The simplest way is just to insert the constraint into the equation itself - this just sets an equality, where the independent variables of interest must equal the constraint. A classic example of this approach is the basic public goods model (see Persson and Tabellini, 2000).

Imagine an economy where individuals earn an income y_i , where income is distributed log-normally - that means it skews rightwards - there are a few *very* rich people and lots of poorer people. Imagine also that people can be taxed to pay for a public good, g which has very powerful positive impacts on utility at small levels but with diminishing returns, through parameter α . Let's establish people's basic utility, u_i when they get the public good without being taxed - a happy, magical world.

$$u_i = y_i + g^\alpha$$

Taking the first derivative of utility with respect to the public good we get:

$$\frac{\partial u_i}{\partial g} = \alpha g^{\alpha-1} > 0$$

So people want an infinite amount of public goods, albeit at a decreasing rate (and if we adapted the equation somewhat we could get a point where the first derivative was zero. But the point is that unconstrained optimization allows us to keep climbing up the function as we please. What if we introduce taxation? Then we get $u_i = (1 - t)y_i + g^\alpha$.

We still haven't set a budget constraint yet though, so we don't know how taxes relate to the public good. Let's do that now. We play a common analytical trick by 'normalizing' the population to one, which means that average income, $\bar{y}$, will equal total income (think about it). Total tax take equals the total income multiplied by the simple tax rate, which equals average income multiplied by the tax rate, which must equal spending on the public good: $t\bar{y} = g$. Now, we can replace the tax term in the equation above with budget constraint: $t = \frac{g}{\bar{y}}$.

$$u_i = (1 - \frac{g}{\bar{y}})y_i + g^\alpha = y_i - g\frac{y_i}{\bar{y}} + g^\alpha$$

With the constraint in place we now look for the 'optimal' level of public goods provision for each individual. Taking the first derivative of utility with respect to public goods:

$$\frac{\partial u_i}{\partial g} = -\frac{y_i}{\bar{y}} + \alpha g^{\alpha-1}$$

This quantity could be greater or less than zero, which means for various levels of public goods, an individual might want more or less of them. In particular, what matters is the *relative* income of individuals vis-à-vis the mean. Note that as people get richer they like public goods less and less. So this is a classic model of redistribution.

Let's now move on to examining the second method of constrained optimization: the Lagrange multiplier method. In this case we can now introduce inequalities as constraints as well as equalities. In this case we have an 'objective' function $f(x_1, x_2)$ that we wish to maximize (or minimize) subject to a constraint $g(x_1, x_2) = c$. We set a Lagrangian equation as:

$$L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda(g(x_1, x_2) - c)$$

Note we have a new variable here, λ , which is known as the *Lagrange multiplier*. We will maximize L with respect to all three variables and take first order conditions (which means setting the first derivative to equal

zero). Let's try this with an actual example. Imagine that you sleep for eight hours a night and thus have sixteen hours to split between work and play. Let's say you get utility from both in diminishing measure - $U = W^{1/2}P^{1/2}$ - but that you have to trade off work and play because of the 'sleep constraint': $W + P = 16$. The simple unconstrained equation has the first derivatives:

$$\frac{\partial U}{\partial W} = \frac{1}{2}W^{-1/2}P^{1/2} > 0 \quad \frac{\partial U}{\partial P} = \frac{1}{2}W^{1/2}P^{-1/2} > 0$$

So, if you never had to sleep - or more accurately, if there were infinite hours in the day - you would work and play as much as possible. But you can't because of the constraint. So now let's form the Lagrangian:

$$L(W, P, \lambda) = W^{1/2}P^{1/2} + \lambda(W + P - 16)$$

Let's take the three partial derivatives of the Lagrangian equation:

$$\frac{\partial L}{\partial W} = \frac{1}{2}W^{-1/2}P^{1/2} + \lambda = 0$$

$$\frac{\partial L}{\partial P} = \frac{1}{2}W^{1/2}P^{-1/2} + \lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = W + P - 16 = 0$$

We now have three equations in three unknowns. Note that the first two equations have elements equal to minus λ , so they must be equal: that is, $\frac{1}{2}W^{-1/2}P^{1/2} = \frac{1}{2}W^{1/2}P^{-1/2}$. This can only be true if $W = P$ - so you work as hard as you play - how boring! How many hours on each, we can figure this out from the final equation, which stipulates that total hours awake are 16. So you work for eight hours and play for eight. Now this may be a mundane example but note that our optimal points are now constrained - in the unconstrained environment we wanted to play and work forever.

Let's also look at λ , which we can calculate from either of the first two equations. λ equals 0.5. This has an extra meaning - it is the boost in utility you would get from having one extra hour awake per day - so it is the derivative of the utility function with respect to the budget constraint. In economics, this is typically thought of as the 'price' of the constraint. Our utility with eight hours of work and eight of play is eight, so it stands to reason that a single extra hour to be split between work and play would

count for one unit of utility. Now you should try and see the split that obtains for the following equation: $U = W^{2/3}P^{1/3}$, or with another budget constraint that $W \geq 4$.

5 Derivatives of Vectors and Matrices

We finish our analysis of optimization by examining how one takes derivatives of vectors and matrices. I don't find Gill to be very transparent on this matter, so let's make this simpler. Let's begin with a vector $\mathbf{x}$, which contains $\{x_1, x_2, x_3\}$, where $y \in \mathbf{R}^1 = f(\mathbf{x}) = f(x_1, x_2, x_3)$. We can use calculus on matrix algebra as we would on scalar algebra.

$$\frac{\partial y}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \frac{\partial y}{\partial x_3} \end{pmatrix}$$

So, we simply take the derivative of y with respect to each element of the vector. This function is often called a *gradient vector*, which is written $\mathbf{D}_x$ or ∇_x . A Gill shows on p.270, we can go further than this and take all the second derivatives of the gradient to produce a matrix called the *Hessian* that is composed of all the second derivatives and cross-derivatives.

$$H = \begin{pmatrix} \frac{\partial^2 y}{\partial x_1^2} & \frac{\partial^2 y}{\partial x_1 \partial x_2} & \frac{\partial^2 y}{\partial x_1 \partial x_3} \\ \frac{\partial^2 y}{\partial x_2 \partial x_1} & \frac{\partial^2 y}{\partial x_2^2} & \frac{\partial^2 y}{\partial x_2 \partial x_3} \\ \frac{\partial^2 y}{\partial x_3 \partial x_1} & \frac{\partial^2 y}{\partial x_3 \partial x_2} & \frac{\partial^2 y}{\partial x_3^2} \end{pmatrix}$$

You will meet this beauty a number of times when doing statistical work, especially maximum likelihood. Just think of it as showing the second derivatives along the diagonal and the cross derivatives in the elements off the diagonal.

In multidimensional space, we can use the gradient vector and the Hessian to find maxima and minima. If $\nabla_x = 0 \quad \forall x \in \mathbf{x}$, then we know that for each element of $\mathbf{x}$ we are at a maximum, minimum, or inflection point. To discover whether we are at a local maximum or minimum we can look at whether the Hessian is positive definite (local maximum), negative definite (local minimum), or indefinite (a saddle-point). For the maximum

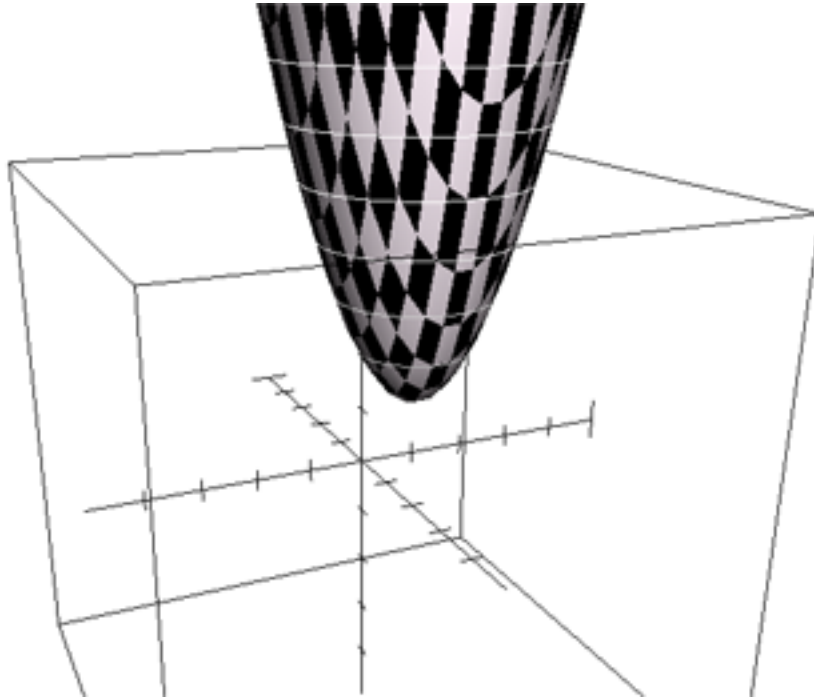


Figure 3: Finding a Minimum Using the Hessian

or minimum to be global, these relationships must hold *for all* x . Let us conclude on a simple two-dimensional example:

$$y = f(\mathbf{x}) = (x_1 - 1)^2 + x_2^2 + 1$$

The gradient vector looks as follows:

$$\nabla_x = \begin{pmatrix} 2(x_1 - 1) \\ 2x_2 \end{pmatrix}$$

And the Hessian is:

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

It is easy to see that this matrix is everywhere positive definite so we know that the function has a global minimum. To find the minimum we look at the first order conditions implied by the gradient vector: $x_1 = 1$; $x_2 = 0$. At this point, y equals one. We can see this graph in Figure 3.